

BOAMP Layer 1 Technical Report

BOAMP-native renewal linkage and survival handoff

Current independent study

End-to-end refresh: 2026-07-27

Abstract

Layer 1, `boamp_only`, is the primary BOAMP-native specification. It uses only identifiers present in BOAMP: valid SIRET, then SIREN, then normalized buyer name. The 2026-07-27 end-to-end run contains 84,623 cleaned notices and 3,380 eligible digital/ICT source contracts. The Layer-1 blocker produces 10,862 candidate pairs over 2,005 sources. At the refreshed balanced threshold 0.343167, the handoff contains 1,003 proxy-renewal events (29.7%) and 2,377 censored sources (70.3%). Median survival is not reached; restricted mean survival time at 60 months is 43.51 months.

1 Scope and Implementation

The source population is Pays de la Loire BOAMP notices from 2015-01 through 2026-07, filtered to digital/ICT contracts by CPV divisions 32, 35, 48, 72 or the configured keyword scope. The current cleaned corpus and Layer-1 artifacts are produced by:

- `notebooks/01_data_engineering_pipeline.ipynb`;
- `notebooks/02_eda.ipynb`;
- `notebooks/03_analysis.ipynb`;
- `data/processed/boamp_only/`;
- `reports/tables/`.

For source notice i and later candidate j , the score is

$$S_{ij} = 0.35s_{\text{text}} + 0.30s_{\text{cpv}} + 0.25s_{\text{time}} + 0.10s_{\text{buyer}}.$$

Only the rank-1 candidate for each source may be accepted. The refreshed Layer-1 rank-1 percentile thresholds are:

$$\tau_{\text{broad}} = 0.278387, \quad \tau_{\text{balanced}} = 0.343167, \quad \tau_{\text{strict}} = 0.442070.$$

2 Current Results

3 Interpretation

Layer 1 is internally coherent and remains the primary specification because it does not depend on external buyer enrichment. Its main limitation is blocking: 1,375 eligible sources never receive a candidate and are structurally censored. The 327 POTENTIAL-tier links are retained in the event set but must be treated as lower-evidence links in sensitivity analysis.

The duration covariate is not causal. Duration is used to construct estimated end dates, temporal blocking, temporal scoring, and the survival covariate itself. The duration-leakage counterfactuals in `notebooks/03_analysis.ipynb` therefore remain part of the evidence standard for any survival interpretation.

Table 1: Layer-1 production outputs from the 2026-07-27 run.

Quantity	Value
Cleaned notices	84,623
Eligible source contracts	3,380
Sources with at least one candidate	2,005
Zero-candidate sources	1,375
Candidate pairs	10,862
Balanced accepted links	1,003
Balanced event rate	29.7%
POTENTIAL-tier links (margin < 0.05)	327
Unique selected candidate notices	741
Maximum candidate reuse multiplicity	6

Table 2: Balanced survival handoff.

Quantity	Value
Rows	3,380
Events	1,003
Censored sources	2,377
Censoring rate	70.3%
Survival at 24 months	0.6908
Median survival	Not reached
RMST at 60 months	43.51 months
Cox hazard ratio for log duration	0.3173
Log-normal AFT concordance	0.6231

Table 3: Model-based and synthetic diagnostics. These are not real ground-truth precision/recall.

Diagnostic	Estimate	Caveat
Fellegi–Sunter precision	0.602	blocked-pair model only
Fellegi–Sunter recall	0.212	conditional on blocked pairs
Combined corruption recovery at R3	0.490	synthetic perturbation
CPV-only corruption recovery at R3	0.767	synthetic perturbation
Text-only corruption recovery at R3	0.845	synthetic perturbation

4 Current Source of Truth

The machine-readable source of truth is:

- `data/processed/boamp_only/boamp_only_sources.csv`;
- `data/processed/boamp_only/boamp_only_candidate_pairs.csv`;
- `data/processed/boamp_only/boamp_only_links_balanced.csv`;
- `data/processed/boamp_only/boamp_only_survival.csv`;
- `reports/tables/linkage_threshold_sweep_by_layer.csv`;
- `reports/tables/survival_layer_conclusions_comparison.csv`.